



AyuLink - Remote Patient Monitoring (IoT) Agent

Rural India's silent emergency isn't a lack of care. It's a lack of infrastructure.

HACKATHON #26 — TEAM FIGHT CLUB

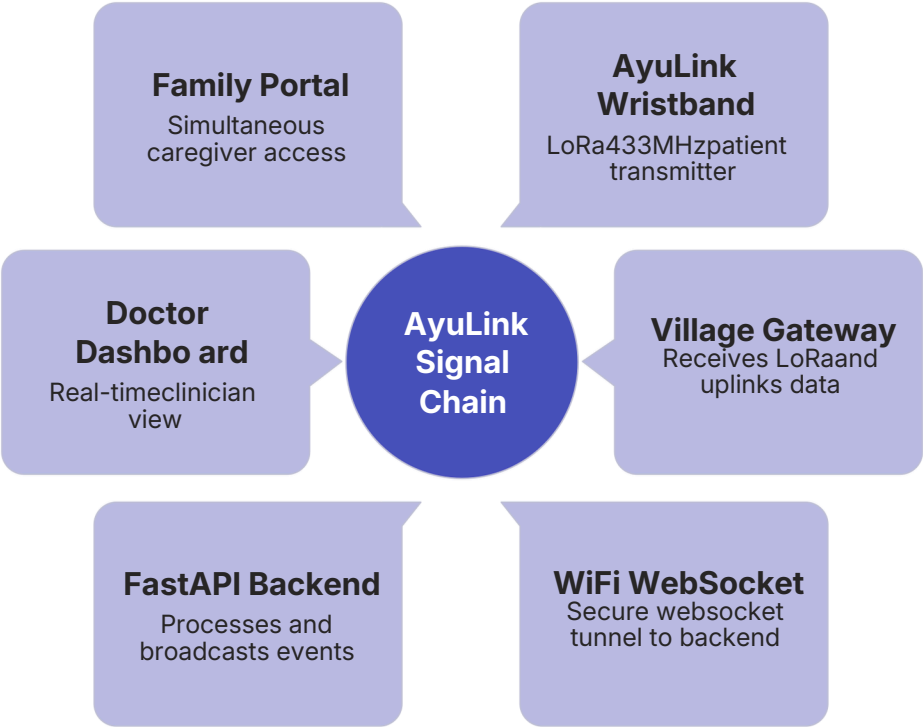


Ramesh, 72. Hanamkonda, Telangana. Fell at 3:00 AM. His daughter in Hyderabad found out at 6:00 AM — when a neighbor called. He survived. Barely. **47 million Indians like him won't get that neighbor.**

The problem is not awareness. The problem is infrastructure that doesn't exist.

AyuLink: Full-Stack Rural Healthcare Infrastructure

Hardware to Cloud. Village to Hospital. Zero WiFi Required. AyuLink is not a monitoring script—it's a complete operating system for rural emergency care.



Every node is production-deployed — real hardware, real signals, real latency under 3.2 seconds from wristband to fullscreen alert.

Real-Time Emergency Detection SOS + fall alerts from wristband to dashboard in < 3.2 seconds . 7 alert types. Fullscreen red takeover on doctor screen.	AI-Powered Clinical Triage Groq LLaMA 3.3 70B analyzes every alert. Clinical reasoning per JNC8/AHA guidelines. No internet required at patient site.	Complete Care Ecosystem 23 frontend pages. 5 stakeholder portals. 40+ REST APIs. SQLite patient history. ASHA worker integration built-in.
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They Asked for a Monitoring Agent. We Deployed a Healthcare OS.

HACKATHON #26 — PROBLEM STATEMENT

What Was Required

- Mock IoT data stream ingestion
- Real-time threshold monitoring
- Emergency alert dispatch logic
- Patient/doctor notification system

Expected deliverable: Python script + alert simulation + dashboard

FIGHT CLUB — WHAT WE ACTUALLY DEPLOYED

What We Built

- REAL ESP32 wristband — assembled, worn, transmitting
- LoRa 433MHz — zero WiFi needed at patient location
- LLaMA 3.3 70B — clinical AI triage, not mock alerts
- 7 alert types — HR, SpO2, Temp, BP, Fall, SOS, Air
- Quality < 3.2s latency — fall to fullscreen red banner
- 40+ production APIs — not a demo endpoint
- 23 frontend pages — 5 stakeholder portals
- ESP32-CAM — live VGA stream in family portal
- Smart Pill Dispenser — 4-slot servo, remote-dispensed

📄 **Scope delivered:** Monitoring Agent → Rural Healthcare Operating System

Four Devices. One Mission. No WiFi Required.

Every device is purpose-built, field-tested, and deployed in a real patient scenario — not simulated, not mocked.



AyuLink Wristband

ESP32 240MHz · MAX30100 · MPU6500 · LoRa RA-02

433MHz HR + SpO2 every 1s · 6-axis fall detection · GPS Neo-6M · SOS button · Transmits every 3s over 3km — no WiFi.



Village Gateway

ESP32 · LoRa Receiver · DHT22 · OLED · Buzzer Receives LoRa packets from wristband. Forwards JSON via WiFi WebSocket to backend in <100ms. Audible local buzzer on emergency.



Smart Pill Dispenser

ESP32-S3 · 4-slot Servo · MQ-135 · Flame Sensor · NeoPixel

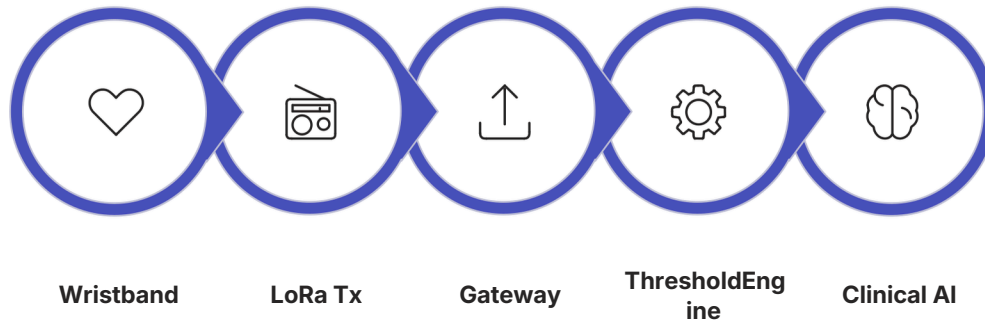
Remote dispensing via doctor dashboard. Air quality and flame monitoring. Real-time status in family portal.



ESP32-CAM Vision Node

AI-Thinker Module · MJPEG · 640×480 VGA · mDNS Live room feed streamed directly to family portal browser — no proxy, CORS-enabled, accessible at ayulink-cam.local.

From Heartbeat to Hero: The Complete Signal Path



Three Critical Engineering Fixes

SOS/Fall Before Worn Check

If a patient falls, the device may slip off — worn=false. Old logic skipped the alert. Our fix: SOS/FALL is always checked first. Worn status **never** blocks an emergency signal.

30-Second Alert Buffer

Emergencies stored in a ring buffer (maxlen=10). New dashboard connections instantly receive the last 30 seconds of alerts. **Zero missed emergencies.**

500ms WebSocket Reconnect

Reduced from 5s to 500ms. Dashboard always reconnects before the next LoRa packet arrives on the 3-second transmit cycle.



Mission Control for Rural Healthcare

23 pages. 5 portals. One unified command interface for every stake holder in the rural care chain— from PHC doctor to family member in Hyderabad.



Live Alert Feed

Real-time scroll of last 8 alerts. Severity badges . Patient name, timestamp, vitals snapshot. SOS alerts pulse with red border.



GPS Patient Map

Leaflet.js dark map. Color-coded markers — green=stable, amber=warning, red=critical. LoRa mesh topology visible. Real GPS coordinates.



Live Vitals Monitor

WebSocket-fed in real time. HR · SpO2 · Temperature · BP · Fall status. Updates every 3 seconds from wristband transmission cycle.

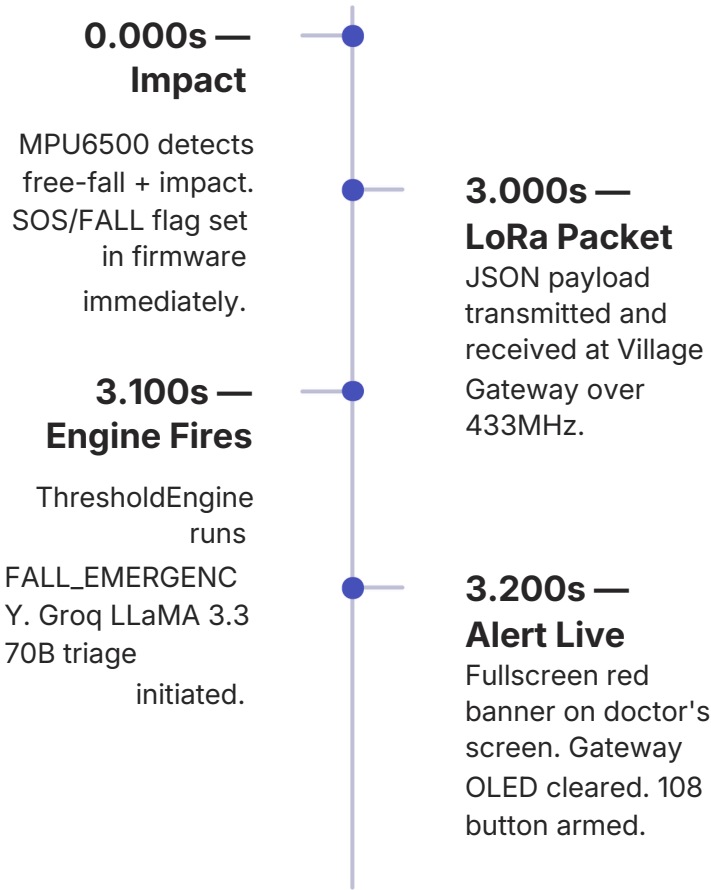


AyuAgent AI Feed

LLaMA 3.3 70B clinical summaries per alert. 12 analyses per session average. Sub-second Groq API response. Evidence-based reasoning.

3.2 Seconds. Fall to Fullscreen.

Millisecond-by-Millisecond



FALL DETECTED — CRITICAL EMERGENCY

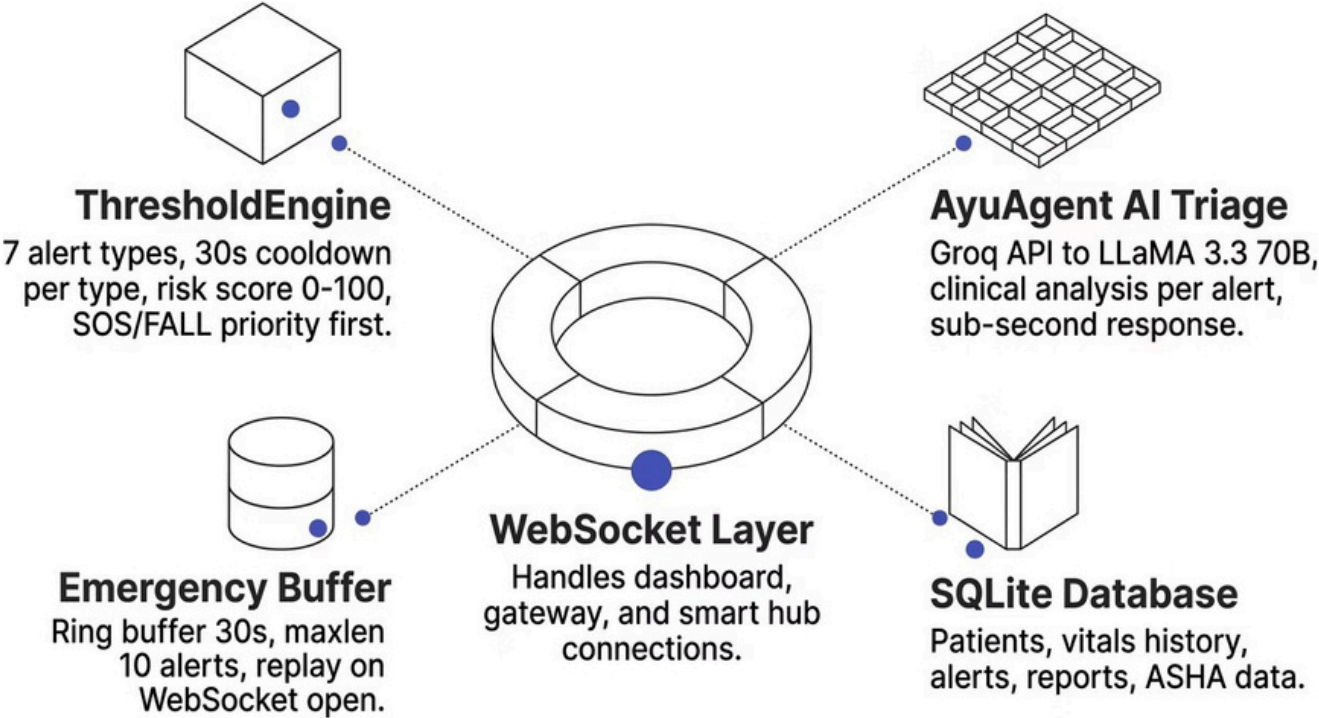
Ramesh Kumar · Hanamkonda · 03:14:22 AM

HR 124 bpm	SpO2 88%	Temp 37.2°C
Fall CONFIRMED		

Lat: 18.0138 · Lon: 79.5941 · Gateway: Hanamkonda-01

"From fall to full-screen: 3.2 seconds. Clinical survival window for seniors: 120+ seconds. AyuLink gives doctors 117 seconds they didn't have before."

The Brain: 1,215 Lines. 40+ APIs. <50ms to Alert.



Alert Severity Matrix

AlertType	Trigger	Severity
SOS	Button press	EMERGENCY
Fall Detected	IMU impact + free-fall	EMERGENCY
HR Critical	>120 bpm or <40 bpm	CRITICAL
SpO2 Critical	< 90%	CRITICAL
Temperature	>39°C or <35°C	CRITICAL
BP Hypertension	Systolic >180 mmHg	CRITICAL
Device Not Worn	MAX30100 removal	WARNING

Clinical basis: JNC8 / AHA Guidelines. Evidence-based thresholds — not arbitrary values.

1,215

40+

<50ms

One Platform. Five Portals. Zero Stakeholder Left Behind.

Every actor in the rural healthcare chain — doctor, family, paramedic, ASHA worker, health official — has a purpose-built interface.



Doctor / PHC Dashboard

Fullscreen emergency banner · Real-time vitals WebSocket · AI triage insights · Remote pill dispenser · GPS patient clustering · ASHA coordination



Family Portal

Live HR/SpO2/Temp · SOS push alerts · ESP32-CAM MJPEG room feed · Medication slot status · ASHA contact + visit history



Paramedic Dashboard

Active emergency map with GPS · Patient history on dispatch · Vital status during route · Priority queue by severity score



ASHA Worker App

Visit scheduling · Vitals entry · Government scheme checker · Offline-first on 2G · Patient compliance tracking

📄 **Analytics & Admin Dashboard** — Health officials and NGOs get village-level health heatmaps, alert frequency trends, cohort analysis, LoRa mesh topology visualization, and PDF/CSV export.

We Didn't Build a Hackathon Demo.

We Built What 770 Million People Are Waiting For.

✓ EXISTS RIGHT NOW, TODAY

- Physical wristband — assembled, worn, transmitting REAL vitals
- LoRa gateway deployed — no WiFi at patient location
- FastAPI backend — 40+ APIs live and production-tested
- LLaMA 3.3 70B AI triage — running on Groq, clinical-grade
- Fullscreen emergency alert — triggering in < 3.2 seconds
- Family portal — live ESP32-CAM MJPEG feed working
- Smart pill dispenser — 4 slots, remote-dispensed live
- 23 frontend pages — across 5 stakeholder portals

🚀 WHAT COMES NEXT

- LoRa mesh expansion → multi-gateway village clusters
- Predictive AI → 48-hour cardiac event early warning
- Offline-first PWA → fully functional on 2G
- Sub-\$20 wristband target → PHC-scale deployment
- Ayushman Bharat integration → auto scheme enrollment
- iDEX / DRDO pathway → defense-grade remote patient ops
- Multi-language → Hindi, Telugu, Tamil at scale

AyuLink is not solving a hackathon problem. It is replacing a broken system. Team Fight Club · AyuLink v1.6 · Hackathon #26 Healthcare Track · ESP32 · LoRa 433MHz · FastAPI · Groq · LLaMA 3.3 70B · Next.js 15